$\frac{N^3}{N_{\mathrm{trg}}^2}\left(\sum_\ell \sqrt{q_\ell}\right)^2$. Together with the requirement $M_V \gtrsim \mathrm{poly}\log N \cdot \frac{N^5}{N_{\mathrm{trg}}^2}\left(\frac{\sum_{\ell\in\mathcal{S}}\sqrt{q_\ell}}{\sum_{\ell\in\mathcal{S}} q_\ell \ell^{-1}}\right)^2$ in Corollary 18 we obtain the assertion. □

D Proof of Proposition 8

We first show that

$$\boldsymbol{q}^* = (q_1^*, q_2^*, \ldots, q_U^*) = \frac{(1, 2, \ldots, N_{\mathrm{trg}}, 0, \ldots, 0)}{Z} \quad (Z = \frac{N_{\mathrm{trg}}(N_{\mathrm{trg}}+1)}{2})$$

satisfies the KKT condition of the LP

$$\mathbb{P}: \begin{cases} \text{minimize} & \sum_{\ell=1}^U q_\ell \ell^2 \\ \text{subject to} & \frac{\max_{\ell=1}^U q_\ell \ell^{-1}}{\sum_{\ell=1}^U q_\ell \ell^{-1}} \le N_{\mathrm{trg}}^{-1} \\ & \sum_{\ell=1}^U q_\ell = 1 \\ & q_1, \ldots, q_U \ge 0 \end{cases},$$

and show its uniqueness.

The KKT condition of $\mathbb{P}$ is

$$\ell^2 + (\lambda_\ell - \sum_{\ell'=1}^U N_{\mathrm{trg}}^{-1}\lambda_{\ell'})\ell^{-1} - \mu_\ell + \nu = 0 \quad (\ell \in [U]), \tag{D.1}$$

$$\lambda_\ell(q_\ell \ell^{-1} - N_{\mathrm{trg}}^{-1}\sum_{\ell'} q_{\ell'}(\ell')^{-1}) = 0 \quad (\ell \in [U]), \tag{D.2}$$

$$\mu_\ell(-q_\ell) = 0 \quad (\ell \in [U]), \tag{D.3}$$

$$q_\ell \ell^{-1} \le N_{\mathrm{trg}}^{-1}\sum_{\ell'=1}^U q_{\ell'}\ell'^{-1} \quad (\ell \in [U]), \tag{D.4}$$

$$q_1 + \ldots + q_U = 1 \quad , \tag{D.5}$$

$$\lambda_\ell \ge 0, \mu_\ell \ge 0 \quad (\ell \in [U]). \tag{D.6}$$

We construct $(\boldsymbol{\lambda} = \{\lambda_\ell\}, \boldsymbol{\mu} = \{\mu_\ell\}, \nu)$ such that $(\boldsymbol{q}, \boldsymbol{\lambda}, \boldsymbol{\mu}, \nu)$ satisfies these conditions: by construction, (D.4) and (D.5) are already satisfied. Here, note that

$$q_\ell \ell^{-1} = \begin{cases} Z^{-1} & (\ell \le N_{\mathrm{trg}}), \\ 0 & (\ell > N_{\mathrm{trg}}). \end{cases}$$

Thus, from (D.2) and (D.3) we have $\lambda_\ell = 0$ $(\ell > N_{\mathrm{trg}})$ and $\mu_\ell = 0$ $(\ell \le N_{\mathrm{trg}})$.

Now it remains to satisfy (D.1), not braking (D.6). For $\ell \in [N_{\mathrm{trg}}]$, (D.1) reduces to the following linear equations:

$$\left(\boldsymbol{I}_{N_{\mathrm{trg}}} - \begin{bmatrix} N_{\mathrm{trg}}^{-1} & \cdots & N_{\mathrm{trg}}^{-1} \\ \vdots & \ddots & \vdots \\ N_{\mathrm{trg}}^{-1} & \cdots & N_{\mathrm{trg}}^{-1} \end{bmatrix}\right)\begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_{N_{\mathrm{trg}}} \end{bmatrix} = -\begin{bmatrix} 1^3 + \nu \cdot 1 \\ \vdots \\ N_{\mathrm{trg}}^3 + \nu \cdot N_{\mathrm{trg}} \end{bmatrix}.$$

Noting that the sum of the all entries in the vector obtained by evaluating left-hand side is zero, we obtain

$$\nu = -\frac{1 + \cdots + N_{\mathrm{trg}}^3}{1 + \cdots + N_{\mathrm{trg}}} = \frac{N_{\mathrm{trg}}(N_{\mathrm{trg}}+1)}{2}.$$

We can also observe $\lambda_1 \geq \cdots \geq \lambda_{N_{\text{trg}}}$ since the right-hand side is decreasing w.r.t the vector index. Since $\boldsymbol{w} = [1, 1, \ldots, 1]^\top$ belongs to the right kernel of $\left(\boldsymbol{I}_{N_{\text{trg}}} - \begin{bmatrix} N_{\text{trg}}^{-1} & \cdots & N_{\text{trg}}^{-1} \\ \vdots & \ddots & \vdots \\ N_{\text{trg}}^{-1} & \cdots & N_{\text{trg}}^{-1} \end{bmatrix}\right)$, we can shift $\boldsymbol{\lambda}$ by this vector to ensure $\lambda_{N_{\text{trg}}} = 1$ (then $\lambda \geq 0$), meaning

$$1 - \bar{\lambda} = -N_{\text{trg}}^3 - \nu \cdot N_{\text{trg}} \Leftrightarrow \bar{\lambda} = \frac{1}{2}N_{\text{trg}}^3 - \frac{1}{2}N_{\text{trg}}^2 + 1$$

where $\bar{\lambda} = N_{\text{trg}}^{-1} \sum_{\ell=1}^{N_{\text{trg}}} \lambda_\ell$. We now consider (D.1) with $\ell > N_{\text{trg}}$. Since

$$\begin{aligned} \mu_\ell &= \nu + \ell^2 - \ell^{-1}\bar{\lambda} \\ &\geq (N_{\text{trg}} + 1)^2 - \frac{N_{\text{trg}}(N_{\text{trg}} + 1)}{2} - \frac{1}{2}N_{\text{trg}}^2 + \frac{1}{2}N_{\text{trg}} - \frac{1}{N_{\text{trg}}} \qquad = 2N_{\text{trg}} + 1 - \frac{1}{N_{\text{trg}}} \geq 1, \end{aligned}$$

and then we can now determine $\boldsymbol{\mu}$ satisfying (D.6). Therefore, obtained $(\boldsymbol{q}^*, \boldsymbol{\lambda}, \boldsymbol{\mu}, \nu)$ satisfies the KKT condition.

From [Man79], to show the uniqueness it suffices to show that for any $\boldsymbol{p} \in \mathbb{R}^U$, there exists $\epsilon > 0$ such that even if we replace the objective function to $\sum_{\ell \in [U]}(\ell^2 + \epsilon p_\ell)q_\ell$, $\boldsymbol{q}^*$ is optimal. We can easily see this by reconsidering KKT condition—while the only effect by changing the objective is the nonnegativeness of $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$ (D.6), these parameters are continuous with respect to the perturbation, and we can still ensure nonnegativeness since for $\boldsymbol{q} = \boldsymbol{0}$ we already obtained positive parameters.

E Detailed Experimental Setting

E.1 Detailed Settings for Section 4.2

We introduce the detailed settings for the full-traning experiment.

Architecture.

- **Embedding.** We use embeddings obtained by concatenating the positional embedding and the token embedding, i.e., $\begin{bmatrix} \boldsymbol{p}_t \\ \boldsymbol{e}_{z_t} \end{bmatrix}$ where $\boldsymbol{p}_t$ and $\boldsymbol{e}_{z_t}$ are one-hot vectors with ones at t-th and z_t-th entries, respectively. The previous-token embedding in (2.2) is omitted.
- **Transformer blocks.** Each layer consists of a single-head attention module with separate Key-Query matrices, a GeLU-based MLP, and residual connections:

  $$\boldsymbol{x}_t \leftarrow \boldsymbol{x}_t + \text{MLP}\big(\boldsymbol{x}_t + \boldsymbol{W}_V \boldsymbol{X}_{1:t} \,\text{Softmax}(\boldsymbol{X}_{1:t}^\top \boldsymbol{W}_K^\top \boldsymbol{W}_Q \boldsymbol{x}_t)\big),$$

  where

  $$\text{MLP}(\boldsymbol{x}) = \boldsymbol{W}_{\text{MLP},2}\text{GeLU}(\boldsymbol{W}_{\text{MLP},1}\boldsymbol{x} + \boldsymbol{b}_1) + \boldsymbol{b}_2.$$

  Three such layers are stacked, followed by a linear projection of size (N, D) that maps the final embeddings (dimension D) to the vocabulary of size N. We initialized $\boldsymbol{W}_K, \boldsymbol{W}_Q, \boldsymbol{W}_V, \boldsymbol{W}_{\text{MLP},1}$ and $\boldsymbol{W}_{\text{MLP},2}$ using Xavier initialization [GB10], while biases $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are initialized from the zero vector. The size of $\boldsymbol{W}_{\text{MLP},1}$ and $\boldsymbol{W}_{\text{MLP},2}$ are $(4D, D)$ and $(D, 4D)$, respectively, where $D = N + L$ is the embedding dimension. The transformed embedding at the last layer is fed into the trainable linear output layer $\boldsymbol{W}^O$ of size (N, D), initialized using Xavier, before softmax.

Training. Training was performed using AdamW with both the learning rate and weight decay set to 10^{-2}, using 32,768 training samples. We prepared 1,024 in-distribution samples drawn from the same distribution as the training data and stopped training once the accuracy exceeded 90% on these samples.